

AirSense AI

Predictive Air Quality Intelligence for Indian Cities

An AI-driven platform that fuses live sensor data, weather signals and city-level pollution trends into forecasts, heatmaps and citizen-ready alerts — moving pollution response from reactive to predictive.



[Live Demo: airsense-ai-rho.vercel.app](https://airsense-ai-rho.vercel.app)

Team Name: Hydra | Team Leader: Anubhav Singh

Built by Anubhav Singh

Pollution Monitoring Is Reactive, Not Predictive



No Early Warning

Cities track pollution after it spikes — residents and authorities react only once air quality has already turned hazardous.



Fragmented Signals

Sensor readings, satellite imagery and weather data live in silos, so no single view forecasts what's coming next.



Citizens Left Uninformed

Vulnerable groups (children, elderly, asthmatics) rarely get timely, personalised alerts before exposure happens.



Limited Policy Tools

Government bodies lack ready analytics to identify hotspots early and target mitigation where it matters most.

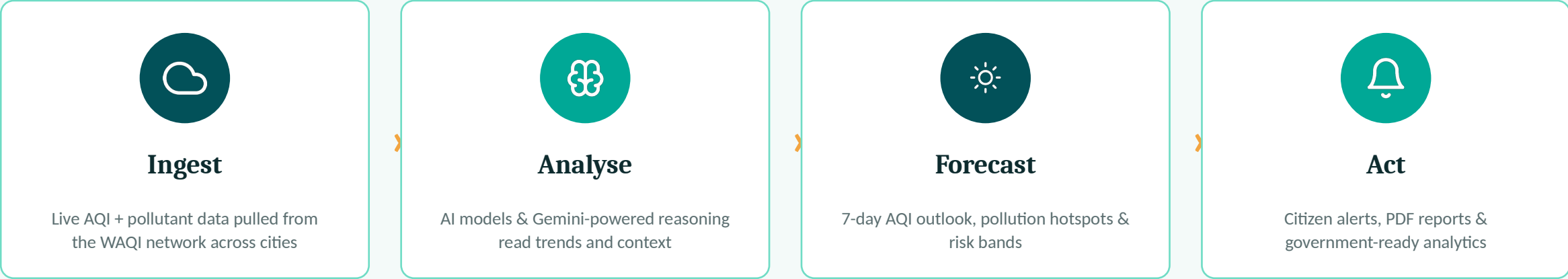
Problem Statement: Develop an AI-driven platform combining satellite imagery, weather information and sensor data to forecast air quality, identify pollution hotspots, and recommend mitigation strategies.

AirSense AI — One Platform, Full Pollution Lifecycle

AirSense AI is a full-stack air quality intelligence platform that turns live monitoring data into forward-looking, actionable intelligence for citizens, businesses and government bodies alike — for 50+ Indian cities today, and expanding.

50+

Indian cities covered
with live AQI data



KEY FEATURES

What AirSense AI Delivers Today



Live AQI Dashboard

Real-time air quality pulled from the WAQI (World Air Quality Index) API across 50+ Indian cities.



7-Day AQI Forecasting

Predictive readouts modelled on historical and meteorological trends.



Pollution Heatmaps

Zone-level PM2.5 / PM10 visualisation, refreshed hourly.



Smart Citizen Alerts

Threshold-based notifications the moment air quality crosses a risk band.



AirBot — AI Chatbot

Google Gemini-powered assistant answering air quality & health queries in natural language.



Downloadable PDF Reports

On-demand pollutant breakdowns with health guidance, generated in-app.



Secure Authentication

Google & GitHub OAuth login via NextAuth.js.



Premium Payments

Razorpay-integrated checkout for premium reports & subscriptions.



Automated Email Delivery

Reports and alerts delivered straight to inbox via Gmail SMTP.

Live, Working Demo — Not Just a Concept

AirSense AI is already deployed and functional. The MVP is live on Vercel with real API integrations — not mockups.



<https://airsense-ai-rho.vercel.app>



Dashboard

City search, live AQI gauge & pollutant breakdown



Forecast & Trends

7-day AQI outlook charted with Recharts



Alerts Centre

Personalised, threshold-based risk notifications



AirBot Chat

Gemini-powered conversational air-quality assistant



Reports

One-click PDF pollutant report generation & email delivery



Account & Billing

OAuth login, profile, and Razorpay premium upgrade

Technology Stack — What's Actually Running



Frontend Framework

Next.js 16 (App Router) · React 19



Styling / UI

Tailwind CSS v4 · Lucide & Simple-Icons icon sets



Backend

Next.js API Routes (Node.js serverless functions)



Database

MongoDB with Mongoose ODM — User, City, Report & Alert models



Authentication

NextAuth.js — Google & GitHub OAuth



AI / GenAI

Google Gemini API (@google/genai) — powers AirBot assistant



External Data API

WAQI (World Air Quality Index) live sensor network



Payments

Razorpay — order creation & signature verification



PDF Generation

@react-pdf/renderer — dynamic pollutant reports



Email Delivery

Nodemailer via Gmail SMTP



Hosting / Deployment

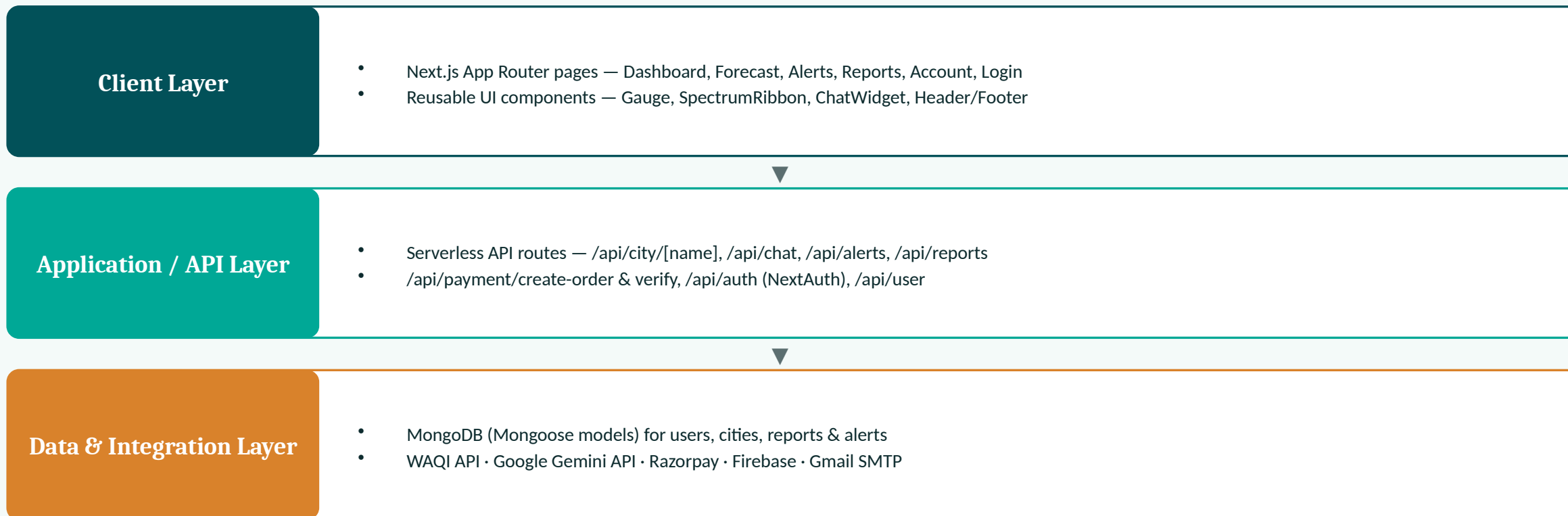
Vercel (CI/CD from GitHub)



Extra Backend Service

Firebase & Firebase Admin SDK

System Architecture



Deployed on Vercel with CI/CD from GitHub • Environment-based secrets for all third-party API keys

What Sets AirSense AI Apart



Conversational AI Layer

AirBot uses Google Gemini to translate raw AQI numbers into plain-language health guidance — not just a chart.



Forecast-First, Not Reactive

7-day predictive outlook shifts the platform from after-the-fact reporting to preventive planning.



Actionable Reports Built In

Auto-generated PDF reports with health recommendations, delivered by email — ready for sharing with authorities or family.



Dual Audience by Design

Same data pipeline serves a simple citizen alert experience and a government-grade analytics view.

Who Benefits From AirSense AI



Everyday Citizens

Real-time AQI, personalised alerts and health tips for daily planning — especially for children, elderly & asthmatic residents.



Government & Municipal Bodies

Hotspot analytics and forecast data to plan targeted mitigation and public advisories.



Urban Planners & NGOs

Zone-level heatmaps to prioritise interventions and track long-term air quality trends.



Businesses & Institutions

Downloadable compliance-ready reports for schools, offices and construction sites.

Built to Grow Beyond the MVP

50+

Indian cities live on the dashboard today

7-Day

Forward-looking AQI forecast window

24/7

Automated monitoring & alerting

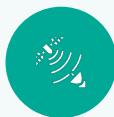
Scalability Path

- Serverless Next.js + MongoDB architecture scales horizontally with city and user growth
- Modular data layer makes it straightforward to add satellite imagery & IoT sensor feeds
- Multi-tenant ready — same platform can license to municipal corporations & enterprises
- Vercel deployment enables instant global edge delivery as adoption expands beyond India

In Active Development — Roadmap Ahead



AirSense AI is a live, working MVP under active development. Core modules — dashboard, forecasting, alerts, AirBot and reports — are functional today, with further refinement and features shipping in upcoming iterations.



Satellite Imagery Integration

Ingest satellite-derived pollution data to strengthen the forecasting model per the problem statement.



Deeper ML Forecasting

Move from trend-based prediction to a trained time-series ML model for higher forecast accuracy.



Mobile App

Native mobile experience for push-based citizen alerts on the go.



Government Analytics Console

Dedicated dashboard for municipal bodies with exportable hotspot analytics.



Hyperlocal Heatmaps

Higher-resolution, street-level pollution mapping using densified sensor + satellite input.



Multi-Country Expansion

Extend the WAQI-based pipeline beyond India to other high-pollution regions.

Thank You

From reactive monitoring to predictive, preventive air quality intelligence.



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Team Members: Aradhya Pandey, Aaradhy Sinha